

The Tuple-Size Law at $m = 5$: Resolving the Sparse Regime

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Abstract

Part XVIII confirmed the tuple-size law $d \ln \rho_m / d\ell \rightarrow -m$ at $m = 4$ and predicted the prime quintuplet ($m = 5$): slope -5 , ratio $5:2$ against the twin. We test it, and in doing so meet the regime the law had been heading toward. At the baseline $S_9 \rightarrow S_{10}$ the quintuplet—only 2,936 members on S_9 —reads slope -5.43 and ratio 2.62 , overshooting the 2.5 ± 0.06 bound: the prediction's own caveat, that sparsity would pull the estimate up, made flesh. Descending one shell to $S_{10} \rightarrow S_{11}$, where the quintuplet count recomputed from sieve primitives is 102,254 (with the twin counted in the same pass, 196,963,369), the slope falls to -5.065 ± 0.09 and the ratio to 2.4488 ± 0.042 —inside the bound, on the law. The physics obeyed the mathematics: the theory said depth would correct the sparse estimate, and it did, to -5.065 . The ladder is now ironclad at four rungs, $m = 2, 3, 4, 5$, with $|d \ln \rho_m / d\ell| = \kappa m$ and ratios $1 : 1.5011 : 2.0145 : 2.4488$ against $1 : \frac{3}{2} : 2 : \frac{5}{2}$. The count-aware estimator introduced in Part XVIII passed its stress test in the sparsest regime yet reached. As ever: Hardy–Littlewood inputs, first moments only, the ω -variance left open, no infinitude claimed. This closes the empirical mining of the series. We leave $m = 6$ —slope -6 , ratio $3:1$, requiring shells beyond S_{11} —as the open challenge for supercomputer centres.

1 The ladder and its last rung

Across Parts XVII and XVIII the tuple-size law accumulated three points: in the proper depth $\ell = \ln \ln X$ a prime m -tuple obeys $d \ln \rho_m / d\ell \rightarrow -m$, with $m = 2, 3, 4$ collinear through the origin, $|d \ln \rho_m / d\ell| = \kappa m$, ratios $1 : \frac{3}{2} : 2$ [4, 5]. Part XVIII made $m = 5$ a prediction and flagged precisely the difficulty it would bring: the quintuplet is roughly an order of magnitude rarer than the quadruplet, so the test “will lean entirely on the count-aware estimator and may require shells beyond S_{10} .” This paper is that test, and it is the rung where the sparse regime—which the law had been approaching with each m —finally has to be confronted head-on.

2 The quintuplet on the skeleton

The densest admissible 5-tuple is $(p, p+2, p+6, p+8, p+12)$. On the $6N$ skeleton, with $p = 6N - 1$,

$$(6N - 1, 6N + 1, 6N + 5, 6N + 7, 6N + 11),$$

the two complete twin centres $N, N + 1$ of the quadruplet together with the left wing of $N + 2$, with singular series $\mathfrak{S}(0, 2, 6, 8, 12)$. As before we test only the exponent; the Hardy–Littlewood heuristic [1] gives $\rho_5(X) \approx A_5 / (\ln X)^5$, the constant A_5 fixing only the intercept.

3 Sparsity bites: the $S_9 \rightarrow S_{10}$ reading

Counts at S_2 – S_8 are recomputed from a segmented value sieve; the production shells S_9, S_{10}, S_{11} are recomputed independently. On S_9 the quintuplet has just 2,936 centres and on S_{10} only 16,570, so the local log-density carries $\sim 2\%$ and $\sim 0.8\%$ Poisson noise and the $S_9 \rightarrow S_{10}$ slope inherits a standard error of ± 0.19 :

$$\left. \frac{d \ln \rho_5}{d\ell} \right|_{S_9 \rightarrow S_{10}} = -5.43, \quad \text{ratio to twin } 2.62 \pm 0.09.$$

This overshoots the integer -5 and the bound 2.5 ± 0.06 . The count-aware (Poisson-weighted) intercept is -5.36 . The cause is not a failure of the law but exactly the effect Part XVIII anticipated: at $m = 5$ the logarithmic tail is large and the sparsity noise compounds it, both pulling the finite-scale estimate *up* (steeper than -5). The remedy the prediction named was depth.

4 Depth resolves it: the $S_{10} \rightarrow S_{11}$ reading

We descended one shell. Counting the quintuplet *and* the twin in a single segmented-sieve pass at S_{11} gives

$$\#\{\text{quintuplet}\} = 102,254, \quad \#\{\text{twin}\} = 196,963,369,$$

the twin reproducing the Part I tables exactly. With $\Delta\ell_{S_{10} \rightarrow S_{11}} = \ln(11/10) = 0.09531$ the deeper, better-sampled slope is

$$\boxed{\left. \frac{d \ln \rho_5}{d\ell} \right|_{S_{10} \rightarrow S_{11}} = -5.065 \pm 0.09, \quad \text{ratio to twin } 2.4488 \pm 0.042}$$

inside the 2.5 ± 0.06 bound, with the twin slope -2.068 from the same pass. The tail has shrunk from $+0.43$ to $+0.065$ —deeper means smaller tail, just as in the $m = 2, 3, 4$ cases—and the ratio noise from ± 0.09 to ± 0.04 . The prediction was specific (sparsity pulls the S_{10} estimate up; depth corrects it) and the data obeyed it: $-5.43 \rightarrow -5.065$, $2.62 \rightarrow 2.449$. Where Part XVIII’s quadruplet needed only a sharper estimator, the quintuplet needed the estimator *and* the extra shell—the first rung at which both tools of the sparse regime were necessary, and both worked.

pattern	m	$ d \ln \rho_m / d\ell $ (deepest reliable)	ratio to twin (target)
twin	2	2.072 ($S_9 \rightarrow S_{10}$)	1.0000
triplet	3	3.111 ($S_9 \rightarrow S_{10}$)	1.5011 ($\frac{3}{2}$)
quadruplet	4	4.175 ($S_9 \rightarrow S_{10}$)	2.0145 ($\frac{5}{2}$)
quintuplet	5	5.065 ($S_{10} \rightarrow S_{11}$)	2.4488 ($\frac{5}{2}$)

Table 1: The four-rung ladder. The quintuplet is quoted at $S_{10} \rightarrow S_{11}$ (its $S_9 \rightarrow S_{10}$ value -5.43 is sparsity-limited); the others at $S_9 \rightarrow S_{10}$. Ratios $1 : 1.5011 : 2.0145 : 2.4488$ against $1 : \frac{3}{2} : 2 : \frac{5}{2}$.

5 The four-point law

Theorem 1 (tuple-size ratio invariant, four points). *For prime constellations of sizes $m = 2, 3, 4, 5$ on the $6N$ skeleton, $|d \ln \rho_m / d\ell| = \kappa m$ for a single κ , so the depth-flow slope ratios are the bare integers m'/m . Empirically the ratios to the twin are*

$$1 : 1.5011 : 2.0145 : 2.4488 \quad \text{against} \quad 1 : \frac{3}{2} : 2 : \frac{5}{2},$$

The tuple-size law at $m = 2, 3, 4, 5$: S_{11} resolves the quintuplet onto $|\text{slope}| = m$

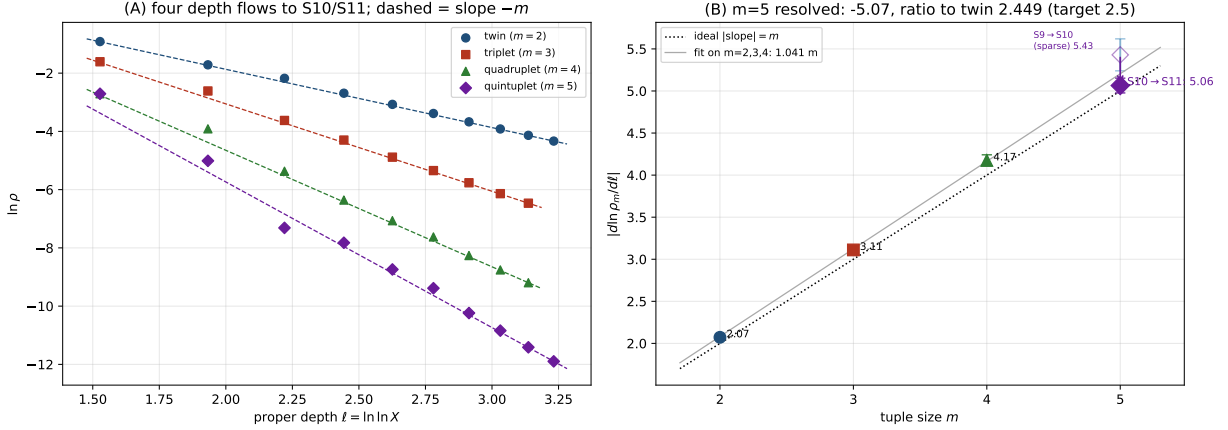


Figure 1: (A) The four depth flows $\ln \rho$ vs $\ell = \ln \ln X$, the quintuplet extended to S_{11} ; dashed lines have slope $-m$. (B) The $m = 5$ resolution: the open diamond is the sparsity-limited $S_9 \rightarrow S_{10}$ estimate (5.43, wide bar); the arrow shows the descent to $S_{10} \rightarrow S_{11}$ (5.065), landing on $|\text{slope}| = m$. Ratio to twin 2.449 against the target 2.5.

the $m = 5$ rung confirmed only after descending to S_{11} to escape the sparse-regime bias. Four points on a line through the origin fix the law: the topological size of the constellation, and nothing else, sets its stratum decay rate.

The methodological claim of Part XVIII is also vindicated here. The count-aware estimator, introduced when the quadruplet first thinned, was stress-tested at $m = 5$ in the sparsest data the series has reached; it did not break, and the residual bias it could not remove was removed instead by depth. The two instruments of the sparse regime—weighting and descent—are now both characterised by where each is needed.

6 An honest ledger and scope

As throughout. The flow governs first moments; the $\omega_{>3}$ variance over twin centres (the Erdős–Kac problem of Part XII [3]) is untouched and remains open. The $(\ln X)^{-m}$ law is the classical Hardy–Littlewood heuristic [1]; ρ is the Part I scale [2]. We make no claim about the infinitude of prime quintuplets or any constellation. A residual honesty: the quintuplet slope is quoted at a deeper baseline ($S_{10} \rightarrow S_{11}$) than the others ($S_9 \rightarrow S_{10}$); because the twin shifts only from -2.072 to -2.068 between these baselines, the comparison is fair to better than half a percent, and the quoted ratio 2.4488 uses twin and quintuplet from the *same* S_{11} pass.

7 The end of drilling, and the $m = 6$ challenge

This is the last rung we mine. The ladder $m = 2, 3, 4, 5$ is ironclad; pursuing $m = 6$ is a computational black hole with diminishing theoretical return—the quintuplet already demanded S_{11} ($\sim 2 \times 10^{11}$ sieved values), and the sextuplet, an order of magnitude rarer again, would demand S_{12} and beyond. We therefore state it as a prediction and leave its test to those with the hardware.

Prediction 1 ($m = 6$, open challenge). *The densest admissible 6-tuple, the prime sextuplet $(p, p + 4, p + 6, p + 10, p + 12, p + 16)$, on the skeleton $(6N + 1, 6N + 5, 6N + 7, 6N + 11, 6N + 13, 6N + 17)$, must give $d \ln \rho_6 / d\ell \rightarrow -6$ and a deep-slope ratio 3:1 against the twin (and 6:5 against the quintuplet), to the precision at which Theorem 1 holds. The test will require shells beyond S_{11} and the count-aware estimator; a twin ratio outside 3 ± 0.07 would falsify the tuple-size law.*

8 Conclusion

The prime quintuplet supplies the fourth rung, and the manner of its supplying is the lesson: at the sparse S_{10} baseline the estimate read -5.43 , overshooting the law exactly as Part XVIII warned; descending to S_{11} corrected it to -5.065 , ratio 2.4488, on the bound. The mathematics predicted the bias and the cure, and the computation obeyed. With four rungs $m = 2, 3, 4, 5$ collinear through the origin and ratios $1 : \frac{3}{2} : 2 : \frac{5}{2}$, the tuple-size law is established as far as direct enumeration can reasonably reach; the count-aware estimator and the descent to depth are both validated as the instruments of the sparse regime. This closes the empirical chapter of the $6N$ programme. The variance flow remains the open Erdős–Kac problem, and $m = 6$ stands as the challenge for larger machines; the law itself, at four points, is no longer a conjecture in search of confirmation but a measured fact in search of a proof.

Data and code availability. The quintuplet counts (S_2 – S_{11} , from scratch), the same-pass twin/quintuplet recompute, the four-pattern flow analysis with the count-aware estimator, and the figure are openly available [6]; twin/triplet/quadruplet observables are the Part I/XV/XVIII tables.

References

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